TECHNICAL BLOG

The 21 percent that actually matters

For a year the public story of test-time compute has been a story about words. o1-class systems spend more decode steps; the answer gets better; the bill grows with the transcript. Snell et al. (2024) made that trade-off quantitative on MATH and GSM8K. BDH-CQ asks a narrower question: **what if the extra work never becomes text?**

Two knobs, not one

Chain-of-thought is a computational workspace that happens to be a language. Coconut (Hao et al., 2024) showed that a Transformer can recycle its last hidden state instead of a token; Geiping et al. (2025) showed that looping a shared block is another way to buy test-time FLOPs without a longer string. HRM and TRM push a different button on ARC: they optimize on the evaluation puzzle — augmentations, identity embeddings, a backward pass — and then vote. BDH-CQ sits in the remaining cell: demonstrations write a recurrent state; the query is solved by iterating a latent workspace; parameters do not move; no CoT is emitted.

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S_t     = U_theta(S_{t-1}, D_t)
H_0     = E_theta(x*, S_K)
H_{r+1} = F_theta(H_r, S_K)
y       = G_theta(H_R)
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Linear attention is the special case S_t = S_{t-1} + U_theta(D_t). Sizes of those maps are withheld. BDH underneath is a separate claim: Hebbian synaptic working memory, sparse non-negative activations, ReLU-low-rank plus linear attention. It is not Mamba.

The number that survived contact with the paper

A 150M configuration scores **29.5% pass@2** on the 400-task public ARC-AGI-1 split at **$0.00070 per task**. pass@1 is 24.25%. An independent black-box audit reproduced 29.5 without weights. Table 5: LOW 21%, MEDIUM 27%, HIGH 29.5%, zero CoT tokens in every row. The paper does not say HIGH means 16 recurrent steps.

Effort	pass@2	Cost vs HIGH
LOW	21%	−22%
MEDIUM	27%	−11%
HIGH	29.5%	0

Two caveats. (1) Table 5 trains at multiple effort levels, then selects at inference. §6.6 MIN vs STANDARD on the deployed system is 111 vs 118 / 400, p = 0.167 — unresolved. (2) Luna is more accurate: 34.2% at $0.040 listed. BDH-CQ's Pareto claim is cost at a given accuracy (57x cheaper at listed price; ~11x after the July cut), not a new accuracy record. HRM ($1.48) and TRM ($1.76) optimize the test puzzle and are off this plane. Snell's MATH curves do not belong on an ARC chart.

What extra R can and cannot do

Dense color permutations: 96/96. Rotation composed with relocation: 72/72. Reflection composed with relocation: 47/72. Color-swap composed with relocation: **0/72**. ConceptARC: FilledNotFilled 9/10, Order 2/10. Pair accuracy 77.9% vs strict-task 59.4%. Generated flood-fill 68.6%; gravity-and-stacking **2.9%**. Latent iteration refines an operator the memory bound. If the operator was never written, waiting is free and useless.

The explainer's live toy is three tasks. Settle: a local fall rule finishes as R grows (caption: the real model is weak at generated gravity). Bind: Hebbian map, done at R = 1. Compose: color map only, never relocates, so R = 4 is as wrong as R = 0. The toy is the linear-attention special case plus a local rule, running in the browser, labeled. It is not BDH-CQ weights.

The tax

Observability. A CoT transcript is a mediocre window, but it is a window. H_R is not. Fixed-size S means interference rather than eviction. Early pretraining 1B–600B is an architectural scaling note, not a second ARC number. SageMaker/AWS is a partnership, not an independent reproduction of the 150M recipe.

What to do with the explainer

Move R on Settle until the grid matches truth. Move R on Compose until you are bored. Then look at Table 5 and say: twenty-one, twenty-seven, twenty-nine point five, seven hundredths of a cent, zero tokens. If HIGH had not beaten LOW on the same split, or if the HIGH point were not cheaper than every plotted system of equal-or-higher accuracy on that August plane, the claim would be false.

References

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